



Optimization of Prestressed Anchor-Cable Support for High and Steep Rock Slopes Using Geological Zoning and Governing Failure Modes

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Abstract: High and steep rock slopes are characterized by pronounced geological heterogeneity, making uniform support parameters prone to either insufficient reinforcement in critical zones or excessive support in high-stability regions. To address this limitation, a geological zoning-based optimization strategy for prestressed anchor-cable support was developed and applied to the Biyinggou quarry slope at the Lawa Hydropower Station, China. The slope was divided into four engineering geological zones according to fault distribution and unloading characteristics, and six representative cross-sections were selected to establish 11 potential sliding modes. Stability analyses were conducted using the simplified Bishop method for circular failure surfaces and the simplified Janbu method for piecewise-linear failure surfaces, while governing cases were verified using the Morgenstern-Price method. Factors of safety (FoS) were evaluated for the natural slope, the unsupported excavated slope, the original support scheme, and the optimized support scheme under persistent, transient, and accidental loading conditions. Among 45 stability scenarios, 12 cases involving five governing failure modes failed to satisfy the design criteria. The first governing failure mode in Section 5-5 and the third governing failure mode in Sections 9-9 and 10-10 consistently exhibited the lowest safety margins. The original prestressed anchor-cable system restored all governing failure modes above the required design thresholds. In high-safety-margin weak unloading zones, the optimized support scheme maintained the required safety performance while reducing anchor-cable density and nominal prestressing intensity per unit slope area. These results demonstrate that support optimization based on engineering geological zoning and governing failure modes can maintain the required safety level while reducing redundant reinforcement, providing an effective and economical strategy for stabilizing large-scale high and steep rock slopes.

Keywords: High and steep slope; Engineering geological zoning; Controlling failure mode; Limit equilibrium method; Prestressed anchor cable; Differentiated support

1 Introduction

Hydropower resources are concentrated in the mountainous canyon region of Southwest China, where large projects often require substantial quantities of natural rock close to the dam site. High, steep slopes in deeply incised valleys are affected by topographic relief, high in situ stress, and tectonic activity. Their stability directly affects material supply, construction platforms, and work areas at lower elevations, and has become a major geological concern for hydropower development in high mountain regions [1]. Quarry slopes face additional disturbance because excavation progressively removes toe support and uses multistage drilling and blasting. Blasting may trigger abrupt displacement along local discontinuities and create a damaged zone of variable thickness near the slope surface [2, 3]. As excavation proceeds, unloading and blast damage accumulate around slope faces and benches and further reduce rock-mass strength [4, 5]. This disturbance is spatially heterogeneous, and its extent depends on the excavation profile, rock-mass quality, and blast-source location [6, 7]. Where the dam-aged zone intersects an adverse joint or fault configuration, local damage can develop into a failure mode that controls overall slope stability [8–10]. Design must therefore determine the overall factors of safety (FoS), the location and geological control of risk, and the support required in each sector.

A credible stability assessment of a rock slope begins with failure modes that represent the actual geological structure. Connectivity among faults, joints, and fractures changes crack-propagation paths and progressive failure. Therefore, a potential sliding mass may be bounded by a steep rear release surface, a gently dipping basal surface, and a shear segment through the rock mass [11–14]. Recent studies have improved non-circular critical-surface search, limit-equilibrium/numerical comparisons, reliability analysis with spatially variable discontinuities, and local FoS evaluation [11, 13, 15–22]. Other studies have emphasized that excavation disturbance ranges and reinforcement forces should be related to the position and evolution of weak structural planes, rather than treated as uniform along a slope [23–25]. These advances support mechanism-based assessment, but they are usually developed for a single slope section, an automatically searched critical surface, or a predefined anchored model. For a high, steep quarry slope where fault assemblages, weathering-unloading boundaries, and excavation geometry change along strike, the remaining challenge is to translate engineering geological zoning into a limited set of controlling failure modes that can guide differentiated support design.

Support design must also account for the response of each controlling mode to different loads. Rainfall infiltration raises pore-water pressure and reduces shear strength along potential slip surfaces, with the response governed by rainfall intensity, duration, and water-induced weakening of the rock mass [26, 27]. Earthquakes add inertial forces and may amplify the adverse effects of existing fractures and weak layers [20, 28, 29]. Recent model tests and numerical studies further show that prestressed anchor cables do not work as a purely uniform stabilizing force: Cable axial force, prestress loss, stress redistribution, and deformation restraint depend on weak-plane geometry, cable spacing, inclination, and the stage of loading [24, 25, 30, 31]. A uniform support specification can therefore produce two design errors: Under-support in sectors containing the most adverse closed sliding mass and over-support in sectors that retain a high safety margin. The scientific contribution required for such projects is not another calculation method alone, but a traceable workflow linking geological zoning, controlling failure-mode identification, multi-condition FoS verification, and support-intensity adjustment.

This study develops such a workflow for the high and steep rock slope at the Biyinggou quarry of the Lawa Hydropower Station. The contributions are threefold. First, the slope is divided into four engineering geological zones using fault boundaries, excavation orientation, and weathering-unloading zones, and 11 candidate failure modes are constructed from six representative sections. Second, each mode is evaluated under persistent, transient rainfall, and accidental seismic conditions with a simplified method appropriate for its slip-surface geometry and with Morgenstern-Price verification, allowing the controlling weak sectors to be identified explicitly. Third, the original anchor-cable system and a reduced scheme in high-margin weak-unloading zones are compared using FoS, deformation, and plastic-zone evidence.

2 Project Setting and Engineering Geology

2.1 Project and Excavation Geometry

The Lawa Hydropower Station is located on the Sichuan-Tibet reach of the upper Jinsha River and forms part of the river's cascade development. The project has an installed capacity of 2000 MW and a concrete-faced rockfill dam with a maximum height of 239 m. The hydraulic complex requires approximately 2.58 million m³ of concrete and shotcrete and 19.52 million m³ of fill. The Biyinggou quarry is planned to supply approximately 13.20 million m³ of usable rock. It lies about 2.8 km upstream of the dam site on the left bank. Ground elevation rises from approximately 2600–2800 m in the gully bottom to about 3780 m at the crest, giving a natural relief of nearly 400 m (Figure 1).

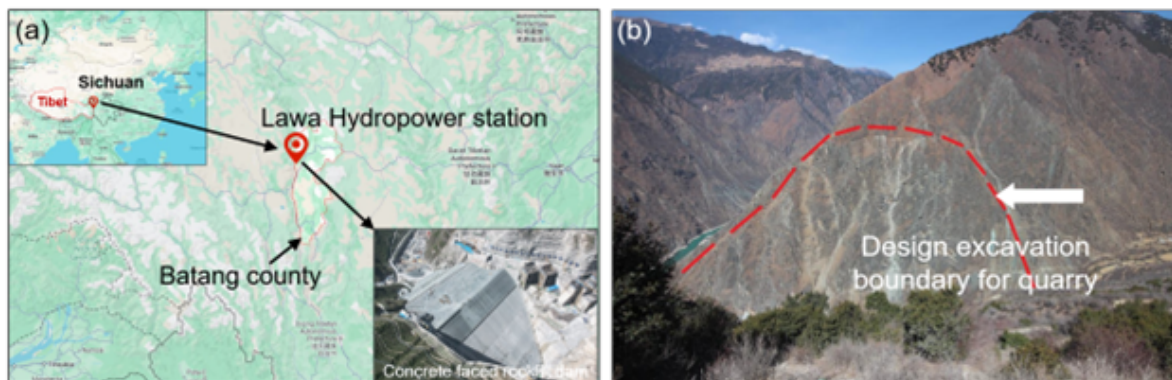


Figure 1. Location of the Lawa Hydropower Station, overview of the Biyinggou quarry slope, and planned excavation extent: (a) geographical location map of the Hydropower Station; (b) a comprehensive view of the quarry slope

The quarry forms a strip along the right bank of Biyinggou, approximately 800 m long and 120–220 m wide. The final excavated slope will be about 360 m high. Slightly weathered to fresh rock is excavated at a near-vertical angle, whereas the slope ratios are approximately 1:0.3 in weakly weathered rock and 1:0.5 in highly weathered rock. A 2-m-wide bench is provided every 20 m of elevation, and an 11-m-wide bench every 60 m.

2.2 Lithology, Weathering, and Unloading Zones

Bedrock is exposed across most of the proposed quarry. Thin residual-colluvial, rockfall-colluvial, and alluvial-proluvial deposits occur only on gentle slopes, at the slope toe, and along the gully floor. The bedrock consists mainly of hornblende schist from the first layer of the lower member of the Proterozoic Xionsong Group, with local mica-quartz schist interbeds. The computational model divides the rock mass into a strongly unloaded and highly weathered zone (V), strongly unloaded and weakly weathered zone (IV₂), weakly unloaded and weakly weathered zone (IV₁), unloading-unaffected and weakly weathered zone (III₂), and unloading-unaffected and slightly weathered zone (III₁). Table 1 lists the natural-state and saturated shear-strength parameters. Saturated parameters were used for the rainfall condition, and natural-state parameters for the other conditions.

Table 1. Rock-mass parameters used in the stability analyses

Stratum	Natural Density (g/cm ³)	Natural Shear Strength Index		Saturated Shear Strength Index	
		f'	c' (MPa)	f'	c' (MPa)
Strongly unloaded-highly weathered (V)	2.0	0.45	0.25	0.38	0.15
Strongly unloaded-weakly weathered (IV ₂)	2.5	0.55	0.40	0.47	0.18
Weakly unloaded-weakly weathered (IV ₁)	2.6	0.70	0.55	0.58	0.30
Unloading-unaffected-weakly weathered (III ₂)	2.8	0.80	0.70	0.68	0.42
Unloading-unaffected-slightly weathered (III ₁)	2.8	0.90	0.85	0.78	0.51

Note: $f' = \tan \varphi'$, in which φ' is internal friction angle; c' is effective cohesion intercept.

2.3 Fault System and Engineering Geological Zoning

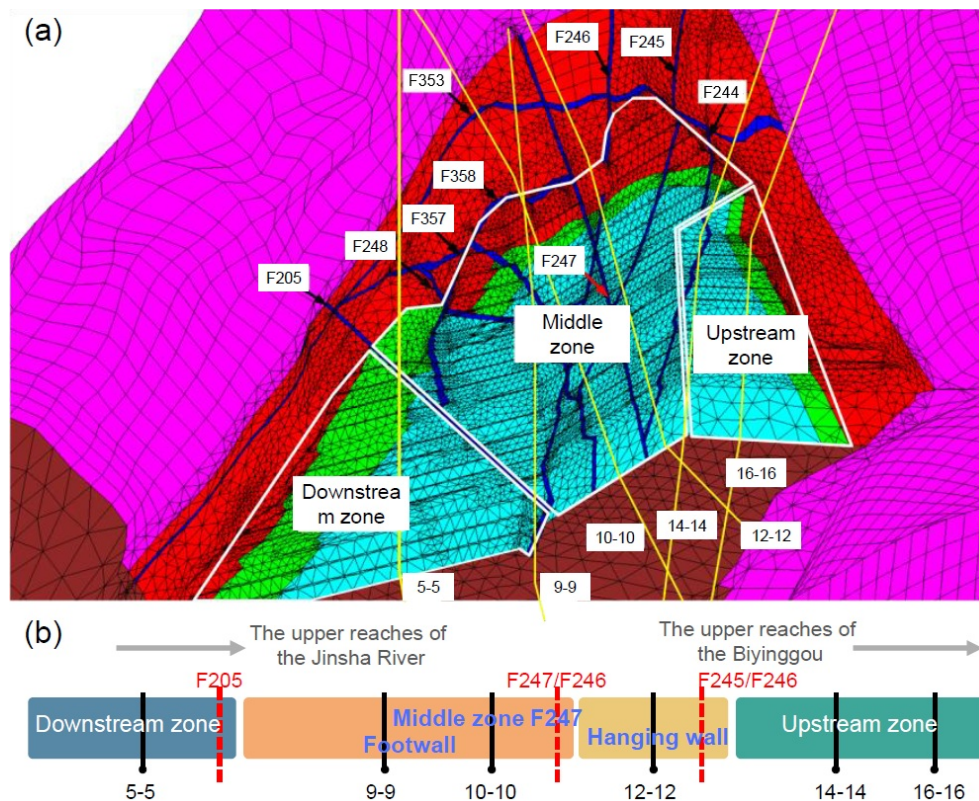


Figure 2. Engineering geological zones: (a) major fault boundaries; (b) six representative sections of the Biyinggou quarry slope

Surface mapping and exploration adits exposed 13 major Class III faults in the quarry. Two dominant sets were identified. The first set generally strikes northwest and dips southwest or northeast at 50°–80°; representative faults include F244, F245, and F246. The second set generally strikes northeast and dips mainly southeast at 70°–80°; representative faults include F354, F355, and F356. The stability analyses also considered F205, F247, F248, F353, F357, and F358. Fault F247 defines the spatial division of the central slope, whereas the F358-F248 configuration controls potential sliding in the footwall zone.

The study area was divided into four engineering geological zones according to the orientation of the excavated slope, fault boundaries, and unloading characteristics (Figure 2). The downstream zone extends from the Jinsha River to Section 6-6 and is represented by Section 5-5. The central F247 footwall zone extends from Sections 6-6 to 11-11 and is represented by Sections 9-9 and 10-10. The central F247 hanging-wall zone extends from Sections 11-11 to 13-13 and is represented by Section 12-12. The upstream zone extends from Sections 13-13 to 17-17 and is represented by Sections 14-14 and 16-16. The resulting two-dimensional analysis system therefore comprised four zones and six representative sections. Table 2 gives the mechanical parameters of the main weak discontinuities.

Table 2. Parameters of the principal weak discontinuities

Discontinuity	Infill Type	Natural Shear Strength Index		Saturated Shear Strength Index	
		f'	c' (MPa)	f'	c' (MPa)
F205, F244, F246, F248, F353, F357, F358	Rock-fragment-filled	0.47	0.15	0.41	0.09
F245, F247	Mud- and rock-fragment-filled	0.32	0.03	0.27	0.02

Note: $f' = \tan \varphi'$, in which φ' is the effective friction angle along the discontinuity surface; c' is the effective cohesion intercept along the discontinuity surface.

3 Methods

3.1 Analysis Framework

The stability analysis began with the engineering geological model rather than an automated search for a single generic slip surface. Faults, weathering and unloading boundaries, and excavation-free faces were first used to define zones and select representative sections. Circular or piecewise-linear slip surfaces were then constructed to match the local geological configuration. Persistent, transient, and accidental load combinations were applied, and FoS values were calculated with both simplified and rigorous methods of slices. Controlling modes were identified where excavation reduced the FoS below the design threshold, after which the original and optimized anchor-cable schemes were checked. Figure 3 summarizes the workflow.

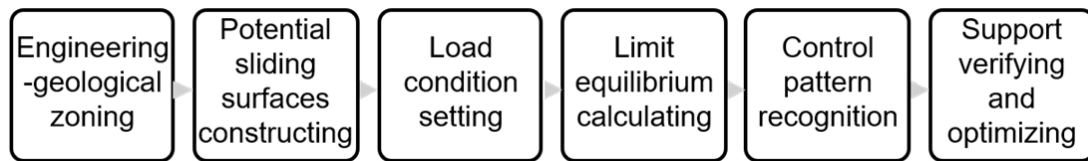


Figure 3. Workflow from engineering geological zoning and failure-mode identification to support verification and optimization

3.2 Identification of Potential Failure Modes

Each potential slip surface comprised a rear tensile-release segment, a basal sliding segment, and a toe daylighting segment. Steep faults that were approximately parallel to the slope or formed a free boundary were preferentially used as rear release surfaces. Candidate basal surfaces were tested in sequence along the lower boundaries of the highly weathered, strongly unloaded, and weakly unloaded zones and along gently dipping faults. Where no continuous basal discontinuity was present, the rear and toe segments were connected by a shear segment through the rock mass. Table 3 lists the potential failure modes in each engineering geological zone, and Figure 4 illustrates the two representative controlling modes. Method selection followed slip-surface geometry: the simplified Bishop method was used for approximately circular surfaces, the simplified Janbu method for piecewise-linear or general non-circular surfaces, and the Morgenstern-Price method for verification.

The candidate modes were screened by three geological and mechanical criteria. First, the rear and basal surfaces had to form a kinematically admissible sliding mass that could daylight toward the excavated face after staged

excavation. Second, the basal surface had to coincide with a persistent weak boundary or a mapped fault where the shear strength was lower than that of the surrounding rock mass; discontinuities that did not connect to the toe or could not close a sliding mass in the two-dimensional section were excluded from the controlling-mode set. Third, for each zone, surfaces were compared from shallow to deep boundaries so that shallow weathering-controlled, strong-unloading-controlled, weak-unloading-controlled, and fault-controlled alternatives were all tested before the lowest FoS mode was identified.

Table 3. Potential failure modes in the four engineering geological zones

Zone/Section	Failure Mode	Rear Surface or Boundary	Basal Sliding Surface	Simplified Method
Downstream/5-5	①	F357	Lower boundary of highly weathered zone	Simplified Bishop
Downstream/5-5	②	F357	Lower boundary of strong unloading zone	Simplified Janbu
Downstream/5-5	③	F357	Lower boundary of weak unloading zone	Simplified Janbu
F247 footwall/9-9, 10-10	①	F358	Lower boundary of highly weathered zone	Simplified Bishop
F247 footwall/9-9, 10-10	②	F358	Lower boundary of strong unloading zone	Simplified Janbu
F247 footwall/9-9, 10-10	③	F358	F248	Simplified Janbu
F247 footwall/9-9, 10-10	④	F358	Lower boundary of weak unloading zone	Simplified Janbu
F247 hanging wall/12-12	①	F353	Lower boundary of strong unloading zone	Simplified Janbu
F247 hanging wall/12-12	②	F353	Lower boundary of weak unloading zone	Simplified Janbu
Upstream/14-14, 16-16	①	Rock-mass shear segment	Lower boundary of strong unloading zone	Simplified Janbu
Upstream/14-14, 16-16	②	Rock-mass shear segment	Lower boundary of weak unloading zone	Simplified Janbu

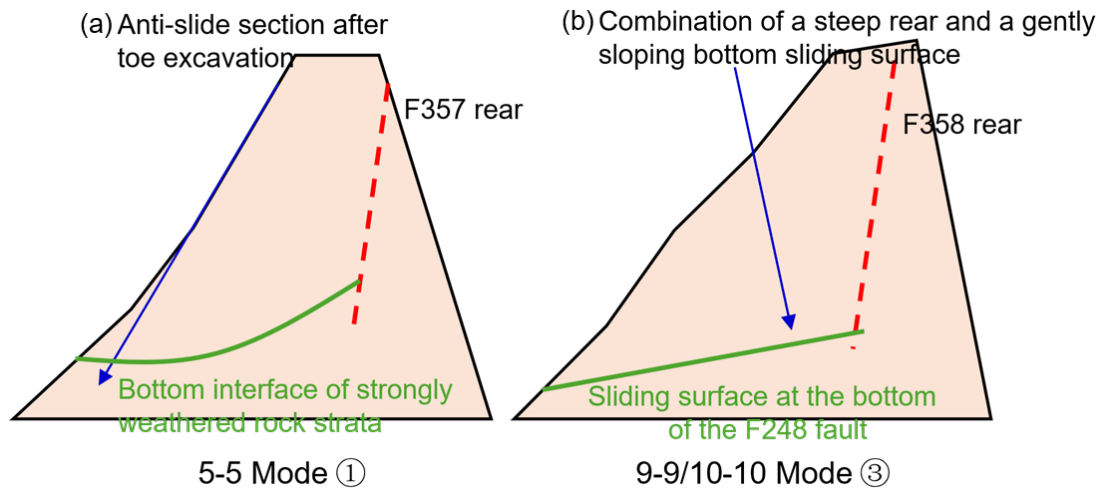


Figure 4. Schematic diagrams of two controlling failure modes: (a) the F357/strongly weathered boundary mode in Section 5-5; (b) the F358-F248 fault-controlled mode in Sections 9-9 and 10-10

3.3 Limit Equilibrium Methods

Slope stability was expressed by the FoS, defined as the ratio of the shear strength available along a potential slip surface to the shear stress required for limit equilibrium. For slice i , the available shear resistance along the slice base is as follows:

$$R_i = c'_i l_i + (N_i - U_i) \tan \varphi'_i \quad (1)$$

where, c'_i and φ'_i are the effective cohesion and friction angle, respectively; l_i is the length of the slice base; and N_i and U_i are the resultant normal force and pore-water force acting on the base, respectively.

The friction coefficient f' listed in the parameter tables satisfies $f' = \tan \varphi'$. The methods of slices differ mainly in their treatment of interslice forces and the overall equilibrium equations.

The simplified Bishop method was used for approximately circular slip surfaces and solved the FoS iteratively from overall moment equilibrium. The simplified Janbu method was used for general slip-surface geometries and was based on overall force equilibrium. The Morgenstern-Price method used $X = \lambda f(x)E$ to describe the relationship between interslice shear force X and normal force E and satisfied overall horizontal force, vertical force, and moment equilibrium [32–34]. For each slip surface, identical parameters and loads were analyzed with a simplified method and the Morgenstern-Price method. The engineering classification was considered insensitive to method choice when both approaches agreed on whether the FoS crossed the design threshold.

3.4 Loading Conditions and Calculation Parameters

The stability analyses included self-weight, measured groundwater or transient rainfall-induced water pressure, prestressed anchor-cable forces, and seismic inertia. The persistent condition included self-weight, groundwater measured during construction, and anchor forces where applicable. The transient condition represented heavy rainfall by assuming a temporarily saturated zone equal to 20% of the potential sliding-mass thickness and by using saturated shear-strength parameters. This engineering assumption was adopted as a conservative simplified representation for a short-duration rainfall event in a fractured rock mass, where transient perched water and strength reduction are expected to occur first near the potential sliding surface rather than throughout the entire slope. The accidental condition added pseudo-static effects for a seismic intensity of VIII to the basic load combination. The horizontal design peak ground acceleration was 0.245 g, the seismic distribution coefficient was 0.25, and the vertical acceleration was two-thirds of the horizontal value. A combination factor of 0.5 was used when horizontal and vertical actions were applied together. These parameters follow the hydropower slope and quarry design framework used for the project and are consistent with pseudo-static screening practice for engineering design [29, 35, 36].

Because the rainfall and seismic representations are simplified design scenarios, their influence was evaluated by comparing the direction and magnitude of FoS changes across the same section-mode combinations. Increasing the assumed saturated thickness or adopting a larger pseudo-static coefficient would reduce the absolute FoS values, whereas decreasing them would increase the calculated margins. The support optimization was therefore not based on a single adverse-case value alone: A reduced scheme was considered only where all three load combinations retained high FoS values and where numerical deformation and plastic-zone results did not indicate a developing deep failure mechanism. Conversely, low-margin modes were kept under the original support even when only one loading condition crossed the design threshold. Table 4 shows the loading conditions and required FoS.

Table 4. Loading conditions and required FoS

Condition	Loads and Water Action	Required FoS
Persistent	Self-weight + measured groundwater; anchor forces included for anchored state	1.20
Transient	Self-weight + transient rainfall-induced water pressure; saturated parameters on slip surface	1.10
Accidental	Basic combination + horizontal and vertical pseudo-static seismic actions	1.05

Note: FoS: factors of safety.

3.5 Slope-Support Schemes and Optimization Criteria

At the excavation crest, the original design used crest-locking anchor-stake, 2000 kN crest anchor cables, and frame beams. Systematic anchor cables in the strong unloading zone were mainly 2000 kN and 40–50 m long, with a horizontal spacing of 5 m and vertical spacing of 4 m. In the weak unloading zone, 30-m-long cables with capacities of 2000 and 1500 kN were arranged alternately on a 5 m grid. Additional crest or spot anchor cables were installed on the upstream dip slope outside the unloading zones and on unloading-unaffected portions of the frontal slope. The present study retained the original support in low-margin sectors and tested a 1500-kN, 6 m by 6 m scheme only in weak unloading zones with high stability margins.

The optimization was constrained by three criteria. First, the FoS had to remain above 1.20, 1.10, and 1.05 under the three respective conditions. Second, the reduction in FoS could not change the stability classification of the failure mode. Third, the numerical analysis had to show limited deformation and no through-going plastic zone. Because a complete cost estimate was not available, support reduction was quantified by anchor-cable density and nominal prestressing intensity. Alternating 2000- and 1500-kN cables in the original weak-zone scheme give a mean prestress of 1750 kN per cable, a density of $1/25 = 0.040$ cables/m², and a nominal prestressing intensity of 70.0 kN/m². The optimized scheme gives $1/36 = 0.0278$ cables/m² and 41.7 kN/m².

4 Results

4.1 Spatial Distribution of Potential Failure Modes

Two basic control mechanisms were identified across the four engineering geological zones. Potential slip surfaces in the downstream and central zones commonly used steep faults as rear release surfaces and followed weathering or unloading boundaries or gently dipping faults at their bases. In the upstream zone, F244 intersects the slope direction at a high angle and F249 is removed by excavation, so neither fault forms a continuous basal sliding surface. Potential sliding there is governed mainly by the lower boundaries of the strong and weak unloading zones. The number of sections was therefore not a measure of risk; the post-excavation stability response depended on the geometry of the rear release surface and basal sliding surface.

Among the 11 failure modes, Mode 1 in Section 5-5 and Mode 3 in Sections 9-9 and 10-10 formed the principal failure paths. The former was bounded by F357 at the rear and the lower boundary of the highly weathered zone at the base, and toe excavation directly removed part of its resisting mass. The latter was enclosed by the steep F358 rear surface and the gently dipping F248 basal fault, forming a wedge-like profile that could slide along discontinuities. Mode 4 in Section 9-9 did not fail under every condition, but its FoS under heavy rainfall fell below the required value and therefore required support.

4.2 Excavation-Induced Changes in Stability

The 15 section-mode combinations produced 45 mode-condition cases for the excavated slope. Twelve cases involving five failure modes did not meet the relevant persistent, transient, or accidental design threshold. Mode 1 in Section 5-5 and Mode 3 in Sections 9-9 and 10-10 were below the threshold under all three conditions and were classified as consistently low-margin modes. Mode 4 in Sections 9-9 and 10-10 fell below 1.10 only under the transient condition and was therefore rainfall-sensitive. The other ten modes retained positive safety margins under all conditions. Figure 5 presents the FoS and margin relative to the applicable threshold for every mode. The low-margin cases were concentrated in a small number of geometrically closed rear/basal configurations rather than distributed continuously along the slope.

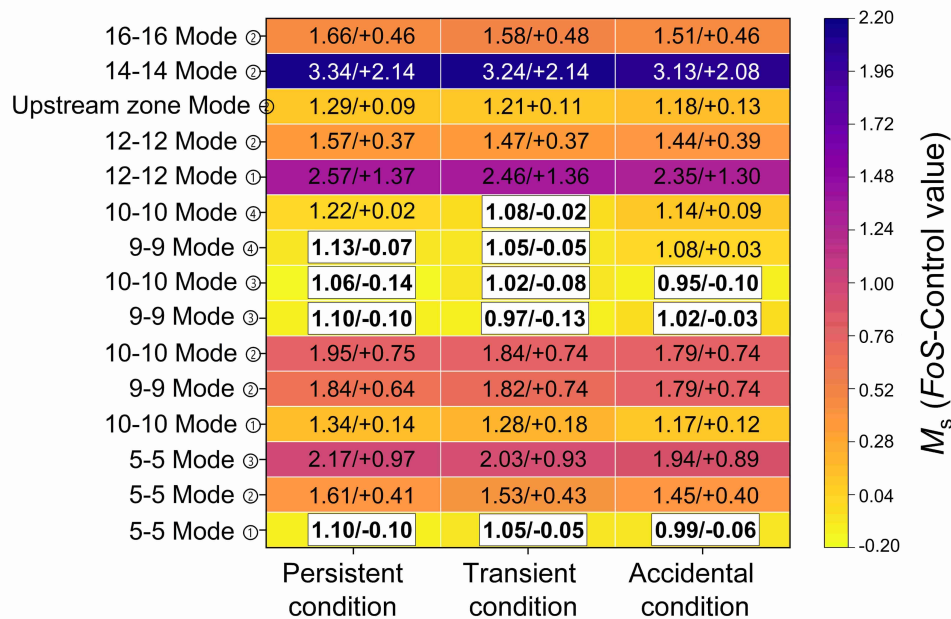


Figure 5. FoS and safety margins for the 15 section-mode combinations after excavation

Note: Black boxes indicate values below the relevant design threshold; the value after the slash in each cell is the FoS minus the threshold; FoS: factors of safety.

Mode 1 in downstream Section 5-5 was most sensitive to toe excavation. The simplified Bishop method gave natural-slope FoS values of 1.67, 1.55, and 1.49, which fell to 1.10, 1.05, and 0.99 after excavation. All three values were below the persistent, transient, and accidental thresholds. The corresponding Morgenstern-Price values were 1.12, 1.06, and 1.00 and led to the same classification. By contrast, Modes 2 and 3 along the lower boundaries of the strong and weak unloading zones retained FoS values of at least 1.45 and 1.94 after excavation. Risk in this zone was therefore concentrated in the shallow to intermediate F357/highly weathered boundary mode rather than along every unloading boundary.

The controlling risk in the central F247 footwall zone was the F358-F248 configuration. For Mode 3, the simplified Janbu method gave post-excitation FoS values of 1.10, 0.97, and 1.02 in Section 9-9 and 1.06, 1.02, and 0.95 in Section 10-10; neither section met the relevant thresholds. Mode 4 had FoS values of 1.13, 1.05, and 1.08 in Section 9-9 and 1.22, 1.08, and 1.14 in Section 10-10, with only the transient rainfall case below the requirement. The central F247 hanging-wall and upstream zones were more stable. Mode 2 in Section 12-12 had FoS values of 1.57, 1.47, and 1.44, and the upstream strong-unloading-boundary mode had values of 1.29, 1.21, and 1.18. The minimum FoS values for the weak-unloading-boundary mode in Sections 14-14 and 16-16 were 3.13 and 1.51, respectively. Figure 6 compares the excavation and anchored states for the main controlling cases.

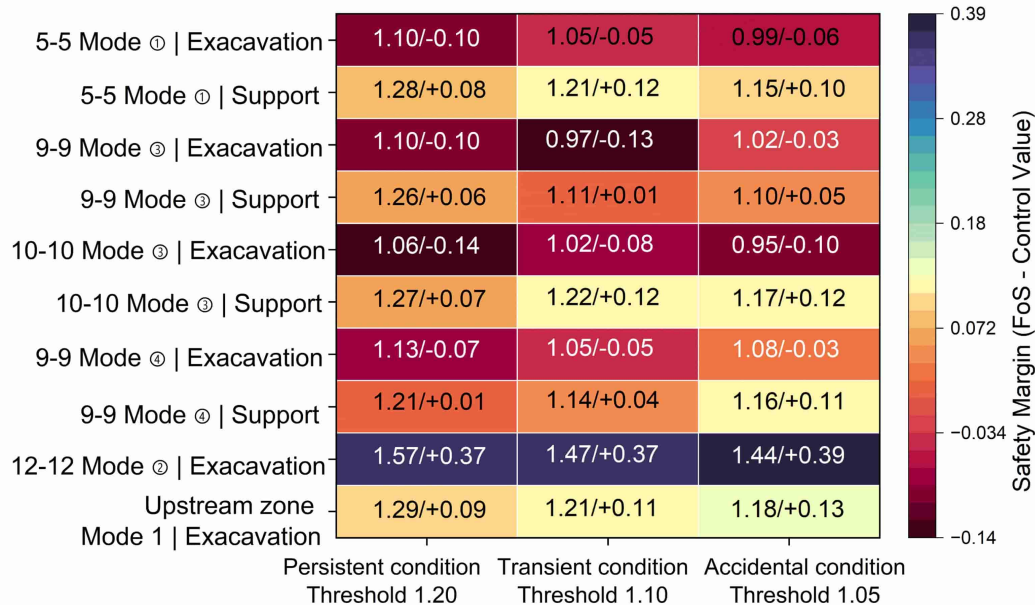


Figure 6. FoS and margins relative to the design threshold for controlling failure modes before and after anchoring
Note: Values in parentheses are FoS minus the threshold for the corresponding condition; FoS: factors of safety.

4.3 Risk Differences Among Loading Conditions

The three load combinations produced different risk patterns. Four, five, and three cases fell below the required values under persistent, transient, and accidental conditions, respectively. Heavy rainfall therefore produced the largest number of non-compliant cases. Across all 15 modes, however, the mean FoS decreased by 0.088 from the persistent to the transient condition and by 0.134 from the persistent to the accidental condition. Seismic inertia thus caused the larger overall reduction. The two metrics describe different effects. Rainfall pushed the near-threshold Mode 4 in Sections 9-9 and 10-10 below the design line, whereas seismic loading caused a broader FoS reduction across most modes, although the accidental-condition threshold was lower.

Load sensitivity also depended on slip-surface material and geometry. Mode 4 followed the lower boundary of the weak unloading zone, so transient saturation and strength reduction acted directly on its main resisting segment and made it critical during heavy rainfall. Mode 3 along F358-F248 already had a low margin under the persistent condition. Its basic weakness therefore arose from geometric closure between a steep rear surface and a gently dipping basal fault, with rainfall and earthquake loading adding to an already adverse structure. Controlling modes should consequently be separated into structurally low-margin modes and modes triggered by a specific load, rather than ranked only by their minimum FoS.

4.4 Consistency Between Limit Equilibrium Methods

For the 90 paired results covering 15 modes, two slope states, and three loading conditions, the simplified Bishop or Janbu method and the Morgenstern-Price method had a coefficient of determination of 0.9996 and a mean absolute difference of 0.019 (Figure 7). The Morgenstern-Price values were generally slightly higher, but the differences did not change the classification of any controlling mode. Both approaches identified the same non-compliant conditions for near-threshold mode 1 in Section 5-5 and Modes 3 and 4 in Sections 9-9 and 10-10. For high-margin modes, the method difference was much smaller than the available safety margin. The spatial risk ranking and support assessment were therefore insensitive to the adopted interslice-force assumption.

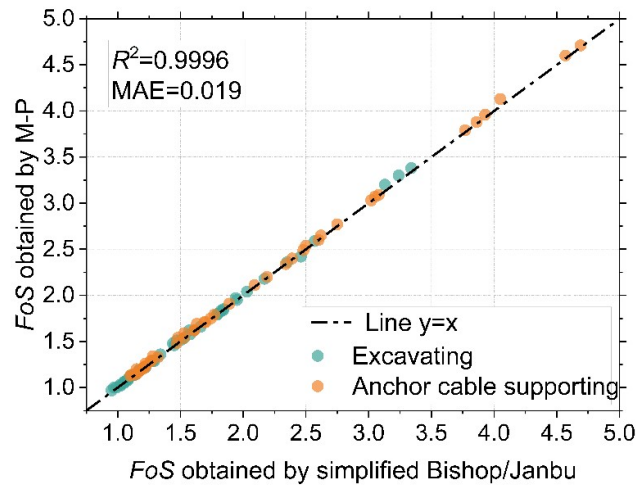


Figure 7. Agreement between results from the simplified Bishop or Janbu method and the Morgenstern-Price method. The dataset contains 15 modes, excavation and anchored states, and three loading conditions

Note: FoS: factors of safety; MAE: mean absolute error.

4.5 Effectiveness of the Original Anchor-Cable System

The original anchor-cable system brought all 45 anchored cases above their applicable design thresholds (Table 5). For Mode 1 in Section 5-5, the simplified Bishop FoS increased from 1.10/1.05/0.99 to 1.28/1.22/1.15, and the Morgenstern-Price values increased to 1.34/1.26/1.20. For Mode 3, the simplified Janbu values increased to 1.26/1.11/1.10 in Section 9-9 and to 1.27/1.22/1.17 in Section 10-10. Under the transient condition, anchoring increased the FoS of Mode 4 to 1.14 in Section 9-9 and 1.20 in Section 10-10, both above the required value.

Table 5. FoS for the main controlling modes after excavation and with the original anchor-cable system (simplified Bishop or Janbu method/Morgenstern-Price method)

Section-Mode	State	Persistent	Transient	Accidental	Assessment
5-5-①	Excavated	1.10/1.12	1.05/1.06	0.99/1.00	Below requirement
5-5-①	Anchored	1.28/1.34	1.22/1.26	1.15/1.20	Meets requirement
9-9-③	Excavated	1.10/1.12	0.97/1.00	1.02/1.03	Below requirement
9-9-③	Anchored	1.26/1.29	1.11/1.14	1.10/1.13	Meets requirement
10-10-③	Excavated	1.06/1.07	1.02/1.02	0.95/0.97	Below requirement
10-10-③	Anchored	1.27/1.29	1.22/1.23	1.17/1.18	Meets requirement
9-9-④	Excavated	1.13/1.14	1.05/1.05	1.08/1.08	Transient case below requirement
9-9-④	Anchored	1.21/1.21	1.14/1.14	1.16/1.16	Meets requirement

Note: FoS: factors of safety.

The gain from anchoring was strongly non-uniform across the slope (Figure 8). The mean Δ FoS for the 45 cases was 0.492, whereas the median was only 0.170, showing that a few large gains influenced the mean. Mean gains under persistent, transient, and accidental conditions were 0.512, 0.501, and 0.463, and the corresponding medians were 0.18, 0.16, and 0.16. Large gains occurred for Mode 1 in Section 10-10, Mode 2 in Sections 9-9 and 10-10, and Mode 2 in Section 14-14, reflecting sensitivity to sliding-mass size, anchorage length beyond the slip surface, and slip-surface position. By comparison, the minimum gains for modes that already had high margins, such as Mode 2 in Section 12-12, were only 0.04-0.06. Anchor performance should therefore be assessed by both threshold recovery and local Δ FoS rather than by a slope-wide average alone.

4.6 Reduced Support in High-Margin Weak Unloading Zones

Support reduction was evaluated only for weak-unloading-boundary modes that retained high safety margins after excavation. For Mode 3 in Section 5-5, the simplified Janbu FoS values were 2.34, 2.19, and 2.09 with the original scheme and 2.29, 2.14, and 2.04 with 1500-kN cables on a 6 m by 6 m grid. The minimum value exceeded the accidental-condition threshold of 1.05 by 0.99. With the optimized scheme, Mode 2 retained FoS values of 4.14, 4.03, and 3.68 in Section 14-14 and 1.80, 1.68, and 1.62 in Section 16-16 (Figure 9). The Morgenstern-Price method gave the same engineering classification.

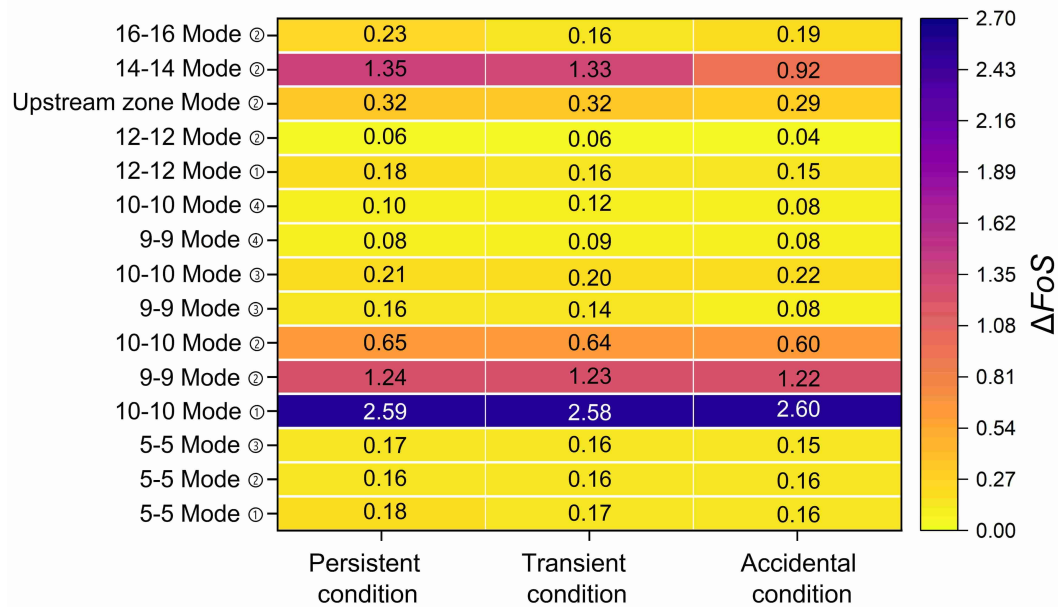


Figure 8. Increase in FoS produced by the original anchor-cable system for the 15 section-mode combinations. Δ FoS is the anchored-state FoS minus the excavated-state FoS

Note: FoS: factors of safety.

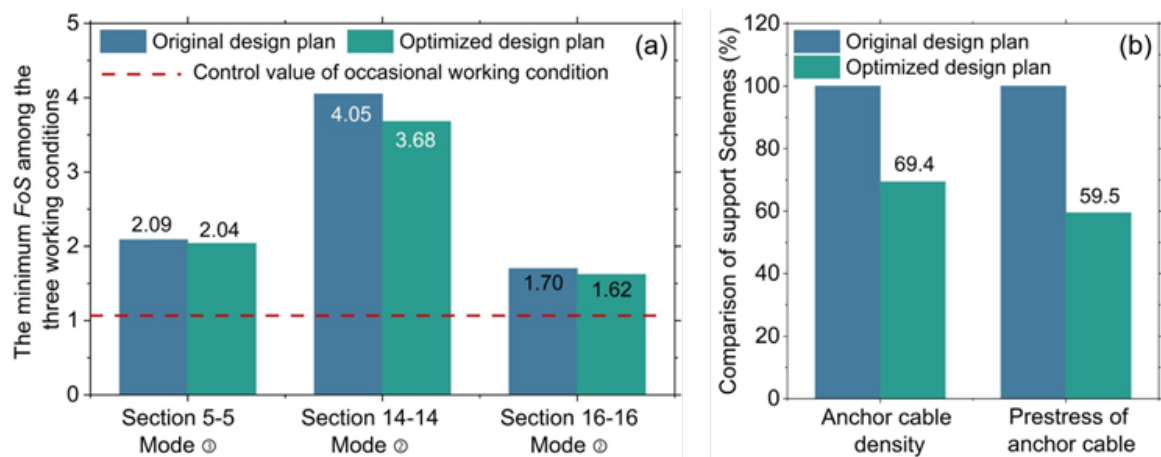


Figure 9. Support optimization in high-margin weak unloading zones: (a) minimum FoS values under the three loading conditions for the original and optimized schemes; (b) relative changes in anchor-cable density and nominal prestressing intensity

Note: FoS: factors of safety.

Table 6. Comparison of the original and optimized schemes in weak unloading zones

Section-Mode	Scheme	Anchor-Cable Parameters	Persistent Factors of Safety (FoS)	Transient FoS	Accidental FoS
5-5-③	Original	Alternating 2000/1500 kN, 5 m by 5 m	2.34	2.19	2.09
5-5-③	Optimized	1500 kN, 6 m by 6 m	2.29	2.14	2.04
14-14-②	Original	Alternating 2000/1500 kN, 5 m by 5 m	4.69	4.57	4.05
14-14-②	Optimized	1500 kN, 6 m by 6 m	4.14	4.03	3.68
16-16-②	Original	Alternating 2000/1500 kN, 5 m by 5 m	1.89	1.74	1.70
16-16-②	Optimized	1500 kN, 6 m by 6 m	1.80	1.68	1.62

Numerical simulations with KBSLOPE indicated that the weak-unloading candidate zones did not develop a deep through-going deformation mechanism. In Section 5-5, the horizontal displacement at the center of the EL2930-2950 m weak-unloading slope segment decreased from 24.9 mm after excavation to 23.5 mm after original support,

while the shear-plastic depth above EL2950 m decreased from 11 m to 0 m. These results indicate that the original anchoring mainly suppressed shallow deformation and local plastic yielding rather than mobilizing a deep sliding body in the weak-unloading optimization zone. Adjusting the grid from 5 m by 5 m with alternating 2000- and 1500-kN cables to 6 m by 6 m with 1500-kN cables reduces the nominal prestressing intensity from 70.0 to 41.7 kN/m², but still leaves all optimized FoS values well above the design thresholds. The reduced scheme should thus be interpreted as a global-stability and deformation-compatible layout for high-margin weak-unloading zones, not as proof that each individual cable has the same utilization ratio as in the original design. Field tension control, lock-off records, and subsequent cable-force monitoring are recommended during implementation to confirm local load redistribution [24, 25, 31]. The reported 30.6% reduction in cable density and 40.5% reduction in nominal prestressing intensity quantify support input per unit area (Figure 9). Table 6 shows the comparison of the original and optimized schemes in weak unloading zones.

5 Discussion

5.1 Influence of Geological Structure on Spatial Stability Differences

Low FoS values at Biyinggou did not vary monotonically with slope height or unloading depth. They were concentrated in geological configurations that enclosed a kinematically admissible sliding mass. For Mode 1 in Section 5-5, toe excavation removed the resisting segment in front of the lower boundary of the highly weathered zone. For Mode 3 in the F247 footwall, the steep F358 fault and gently dipping F248 fault formed a rear/basal configuration in two dimensions. Section 12-12 and the upstream sections were also subjected to high, steep excavation, but lacked a continuous adverse basal discontinuity and therefore retained higher FoS values. This interpretation agrees with evidence that discontinuity connectivity, spatial distribution, and rock-bridge failure jointly govern progressive failure in rock slopes [11–14]. Engineering geological zoning should therefore be used to identify failure mechanisms, not merely to describe site conditions.

Rainfall and earthquakes affected the failure modes differently. Mode 4 in Sections 9-9 and 10-10, which followed the lower boundary of the weak unloading zone, approached or crossed the design threshold mainly during heavy rainfall. Transient water pressure and saturated-strength reduction were therefore its principal adverse factors. By contrast, the F358-F248 Mode 3 had low margins under all three conditions, indicating that the discontinuity geometry itself created the basic weakness. Previous studies have shown that rainfall infiltration, pore-pressure rise, and water-induced weakening can rapidly erode the margin of a critical slip surface, whereas earthquakes tend to increase overall driving forces through inertia and pre-existing damage [20, 26–29]. The present results also show why risk cannot be summarized by the minimum FoS under a single adverse condition: Rainfall produced more non-compliant cases, whereas seismic loading caused the larger mean reduction across all modes. Weak-sector identification should therefore consider the minimum FoS, the number of non-compliant conditions, the mean reduction, and the continuity of the controlling discontinuities.

5.2 Implications for Anchor-Cable Optimization

A uniform support specification extends the demand of the most adverse failure mode across the entire slope. The results support a two-level design strategy. The original support should be retained for Mode 1 in Section 5-5 and Modes 3 and 4 in Sections 9-9 and 10-10 because these modes crossed the design thresholds after excavation or during heavy rainfall. Anchor-cable density and capacity may be reduced in zones governed by the weak unloading boundary where the excavated slope retains a high FoS, provided that the reduced scheme passes multi-condition checks and deformation constraints. The difference between the mean Δ FoS of 0.492 and median Δ FoS of 0.170 across the 45 cases further demonstrates that support effects were spatially non-uniform. This strategy does not replace local risk with an average FoS; it assigns support intensity to the controlling mode in each zone. The approach is consistent with previous work showing that anchor-cable parameters should be selected in relation to fractures, potential slip surfaces, and local safety reserve [30].

The engineering value of the optimized scheme is first expressed as a transparent reduction in support input. Anchor-cable density and nominal prestressing intensity can be calculated directly from design parameters and represent drilling demand and prestressing capacity per unit slope area. Final cost, however, also depends on cable materials, construction platforms, borehole depth, grouting, and schedule. The reported reductions of 30.6% and 40.5% should therefore be interpreted as per-area support indicators.

5.3 Inspiration for Excavation of Other High and Steep Slopes

The transferable outcome is the analysis workflow rather than the site-specific strength parameters or anchor-cable specifications. For another high and steep rock quarry or hydropower slope, the first adjustable component is the geological zoning rule: The boundaries, may be major faults, lithological contacts, bedding orientation, weathering unloading zones, or excavation directions, depending on which features control sliding-mass closure. The second component is the candidate-mode matrix, which should be rebuilt from the local combination of rear-release surfaces,

basal weak layers, rock bridges, and daylighting conditions. The third component is the load scenario: Rainfall infiltration depth, groundwater level, seismic coefficient, and strength reduction should be selected from local codes, hydrological conditions, and seismic hazard levels. The fourth component is the support scheme, for which cable length, inclination, spacing, lock-off load, and anchorage depth must be recalculated against the controlling mode in each zone rather than copied from Biyinggou. This workflow combines experience from non-circular slip-surface searches, limit-equilibrium/numerical comparisons, spatial uncertainty studies, and recent work on anchor-force evolution and weak-plane-controlled anchored slopes [11–13, 15–19, 21–25, 31].

5.4 Scope of Application and Limitations

The method is most suitable for high, steep rock slopes where (i) the main discontinuities can be mapped with sufficient confidence, (ii) potential sliding masses can be represented by two-dimensional sections for design screening, and (iii) anchor cables are used to improve the safety margin of structurally controlled failure modes. Its use is less direct for slopes dominated by strongly three-dimensional wedge release, toppling, deep-seated creep, soil-like flow, or rapidly changing groundwater conditions. The main assumptions are deterministic strength parameters, simplified transient saturation, pseudo-static seismic loading, and design-level anchor forces. The reduced support scheme in this study was not independently recalculated with a detailed cable axial-force model, and no long-term field monitoring data are yet available to verify post-construction cable utilization. These limitations define the boundary of the proposed workflow and indicate where future probabilistic, three-dimensional, hydro-mechanical, and monitoring-based analyses should be added.

6 Conclusions

The main findings of this study are as follows:

(i) Based on fault boundaries, excavation orientation, and weathering and unloading zones, the Biyinggou quarry slope was divided into four engineering geological zones: The downstream zone, central F247 footwall zone, central F247 hanging-wall zone, and upstream zone. Eleven potential failure modes were identified across six representative sections. Stability differences were controlled primarily by geometrically closed rear and basal surfaces rather than slope height alone.

(ii) Twelve of the 45 excavated-slope cases for the 15 section-mode combinations fell below the relevant thresholds and involved five failure modes. The principal low-margin modes were the F357/highly weathered boundary mode in Section 5-5 and the F358-F248 fault-controlled mode in Sections 9-9 and 10-10. Mode 4 in Sections 9-9 and 10-10 was critical under heavy rainfall. The transient condition produced the largest number of non-compliant cases, whereas the accidental condition reduced the mean FoS across all modes by 0.134, showing that the two adverse loads produced different risk patterns.

(iii) The original anchor-cable system brought all 45 anchored cases above the applicable design thresholds. The mean and median FoS increases were 0.492 and 0.170, respectively, confirming a strongly non-uniform reinforcement response. Across 90 paired results, the simplified Bishop or Janbu method and the Morgenstern-Price method gave a coefficient of determination of 0.9996 and a mean absolute difference of 0.019. Method choice did not alter the engineering classification of the failure modes or the assessed effectiveness of anchoring.

(iv) For the high-margin weak-unloading modes in Sections 5-5, 14-14, and 16-16, 1500-kN anchor cables on a 6 m by 6 m grid retained minimum FoS values of 2.04, 3.68, and 1.62, respectively, across the three loading conditions. The scheme reduced anchor-cable density by 30.6% and nominal prestressing intensity by 40.5% per unit slope area. This optimization is applicable to high-margin weak-unloading zones verified by FoS and deformation constraints; low-margin fault- or weathering-boundary-controlled modes should retain the original support or be checked with more detailed reinforcement analysis.

Author Contributions

Conceptualization, Z.X.H.; methodology, W.H.W. and Q.H.J.; software, M.Z. and Y.W.; resources, W.H.W.; data curation, Z.X.H.; writing—original draft preparation, Z.X.H.; writing—review and editing, Z.X.H.; supervision, M.T.; funding acquisition, Z.X.H. and M.T. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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